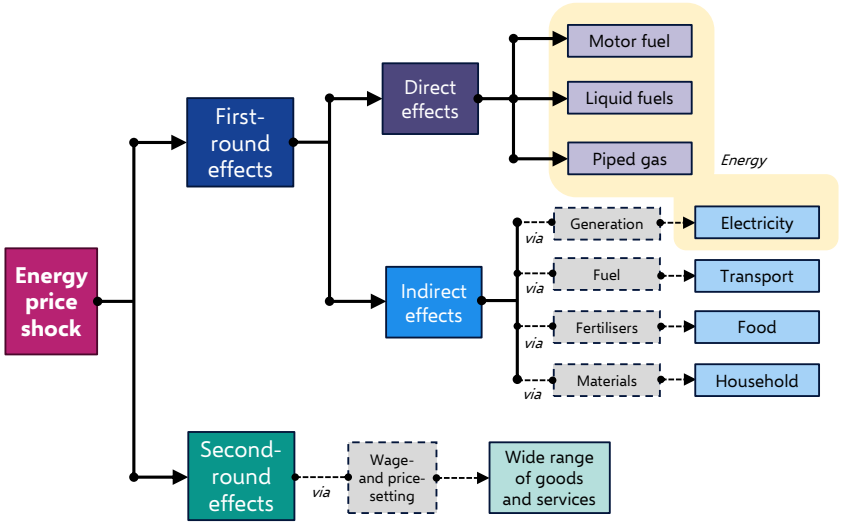
The indirect inflation effects of higher energy prices

- **Beyond their direct effects on consumer prices of fuels and piped gas, higher crude oil and natural gas prices will prompt firms to pass on higher fuel costs to consumers, boosting the prices of non-fuel items of the CPI such as electricity and food. In our baseline scenario, we estimate that these indirect effects on inflation – which are separate from "second-round" effects via broader wage- and price-setting – could range from 0.7%-pt in the US to nearly 1.5%-pt in the euro-zone, with the UK and Japan in between.**
- Chart 1 provides a framework for thinking about how a surge in oil and (natural) gas prices feed through to consumer prices. "First-round" effects are one-off price adjustments to higher fuel costs. "Second-round" effects capture workers and firms responding to first-round price increases by bidding up wages and prices to defend their real incomes. On their own, the first-round effects are (for want of a better word) transitory, even if they can take time to fully feed through. But when inflation expectations rise amid conditions of excess demand, would-be transitory price effects can become self-sustaining via second-round effects.

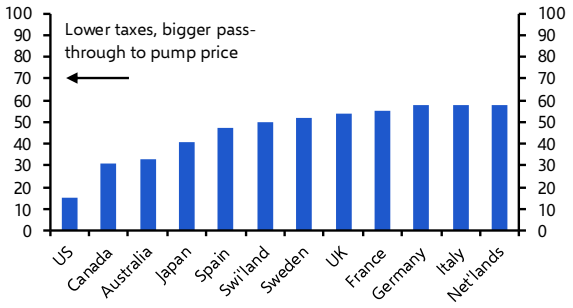
Chart 1: Stylised Transmission of Higher Energy Prices to Consumer Price Inflation



Source: Capital Economics

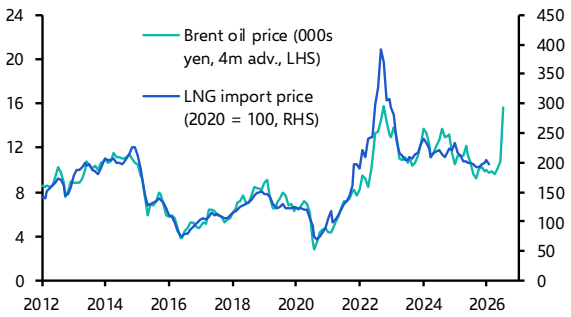
- **Within first-round effects, higher oil and gas prices *directly* push up the consumer prices of fuels.** For motor fuel, the passthrough is bigger where fuel taxes are lower, as in the US. (See Chart 2.) In contrast, in Europe, taxes make up a bigger share of pump prices, which dampens the inflationary effect of a rise in oil prices. Of course, in practice, tax changes or price caps (like those in Japan) will influence the direct passthrough.

Chart 2: Fuel Tax Share of Gasoline Price (%, 2024)



Sources: IEA, Capital Economics

Chart 3: Brent Oil Price & Japan LNG Import Price



Sources: LSEG, Capital Economics

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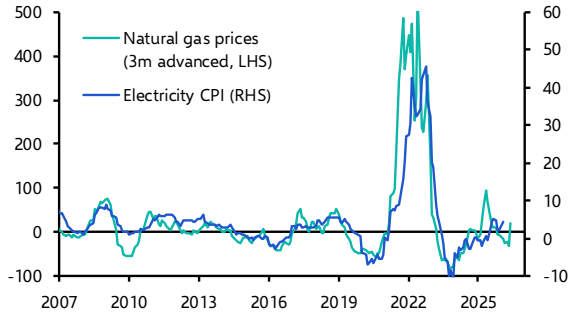
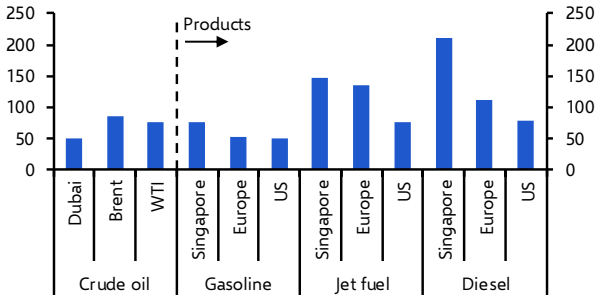
- For natural gas prices, there is also a fairly mechanical passthrough to consumer prices of piped gas, at least outside Japan. Indeed, in Japan, it is the yen price of crude oil that determines the prices of household gas (see Chart 3), not Asian LNG benchmarks, since most imported LNG contracts there are linked to oil prices.
- **In addition to pushing up consumer prices of fuels, higher oil and gas prices *indirectly* boost the prices of non-fuel goods and services as firms pass on higher fuel costs. For example, natural gas prices are a major determinant of electricity prices in many DMs.** (See Chart 4.) In many cases, gas is either a key input for power generation or frequently sets the price of power generated under uniform marginal pricing systems. And as electricity bills are included in some rental agreements, higher gas prices can end up boosting rents.
- **Moreover, higher oil prices increase fuel costs for transport services such as airlines and haulage.** The cost hike for transport services is larger when the cost of higher-grade refined petroleum rises by more than crude oil prices, as is the case today. Indeed, since the start of the war, gasoline prices have experienced comparable increases to crude oil. But the prices of diesel – which is used disproportionately by heavy-goods vehicles transporting goods – and jet fuel have risen much further. (See Chart 5.) Prices of tourism-related services can be sensitive to higher transport costs too.

Chart 4: Natural Gas Prices & Electricity CPI (% y/y, DM Average)



Sources: LSEG, Capital Economics

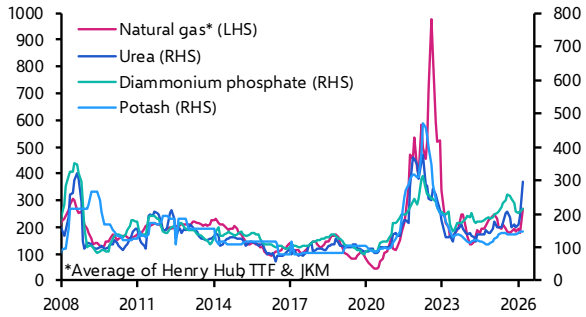
Chart 5: % Change in Prices of Crude Oil & Refined Products Since the War Began



Sources: LSEG, Capital Economics

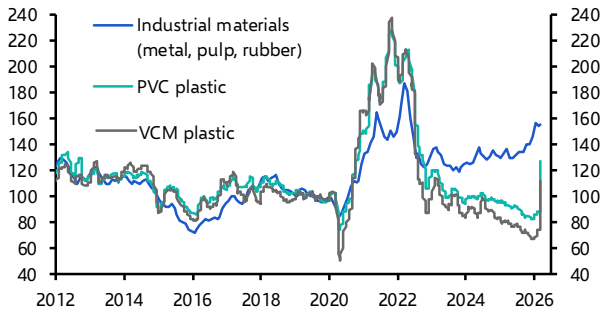
- **Besides electricity and transportation, food production is often energy-intensive, especially in relation to fertilisers.** Natural gas is used to make ammonia-based fertilisers such as urea. While there are gas-free phosphate and potash alternatives, they are partly fungible with and often complementary to ammonia fertilisers, so fertiliser prices are typically jointly driven by natural gas prices. (See Chart 6.) So higher natural gas prices push up food inflation, which in turn increases price pressures in the hospitality sectors.
- **What's more, an energy price shock can increase the costs of production of energy-intensive materials that cascade down supply chains to household goods.** For instance, refined metals, processed rubber, and wood pulp are energy-intensive and hence their prices are significantly driven by energy prices. In addition, higher prices of oil and gas boost the prices of hydrocarbon derivatives such as plastics. The prices of PVC and VCM plastics have spiked since the war began. (See Chart 7.) And given that packaging makes up a higher share of the value of food than other goods, higher plastic prices can exacerbate food inflation.

Chart 6: Natural gas & Fertiliser Prices (2019 = 100)



Sources: LSEG, World Bank, Capital Economics

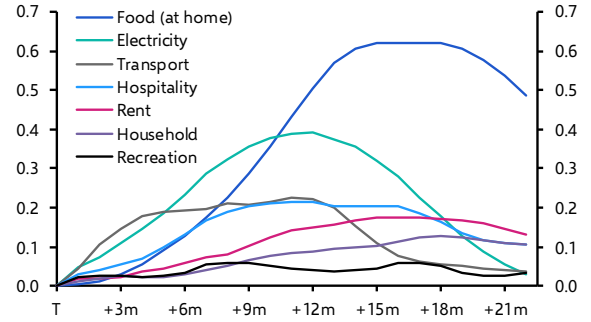
Chart 7: Global Industrial Materials Prices (2019 = 100)



Sources: LSEG, Capital Economics

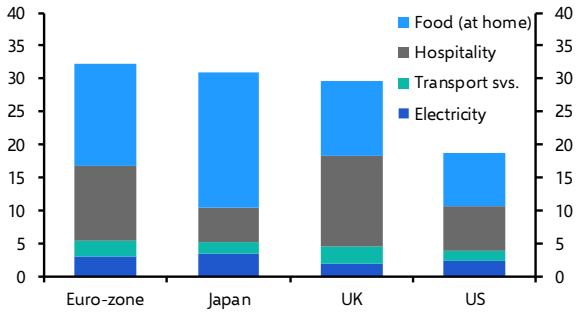
- **Out of all these channels through which a surge in oil and gas prices can boost inflation indirectly, the most important quantitatively is food, followed by electricity.** To determine this, we examine[1] the responses of CPI components to a uniform doubling of oil and gas prices across major advanced economies. In this case, we estimate that the increase in average headline inflation from higher food prices peaks at 0.6%-pt after 15 months, having taken several months to feed through initially. (See the dark blue line in Chart 8.)
- While electricity inflation rises much further than inflation in other CPI components, its small weight in CPI baskets (< 2%) means its contribution to headline inflation is just over half that of food. (Compare the green and dark blue lines.) The inflation boost from higher transport prices is the fastest indirect effect with almost the peak effect coming through in just four months. The peak effect from higher hospitality inflation is similar in size to that of transport but takes much longer to feed through. (Compare the light blue and grey lines.)
- Of course, in current circumstances, advanced economies are not facing a uniform energy price shock. Oil and gas prices have risen by less in the US than in Europe and Asia. And as we set out in our "baseline" and "adverse" energy price scenarios in our *Global Economic Outlook*, this disparity is likely to endure in either case, limiting the indirect boost to inflation in the US compared to other DMs. What's more, the parts of the CPI that are the main transmission channels for indirect energy effects are smaller in the US. (See Chart 9.)

Chart 8: Modelled %-Point Indirect Boost to Inflation from Doubling of Oil & Gas Prices (Major DM Average)



Sources: LSEG, Capital Economics

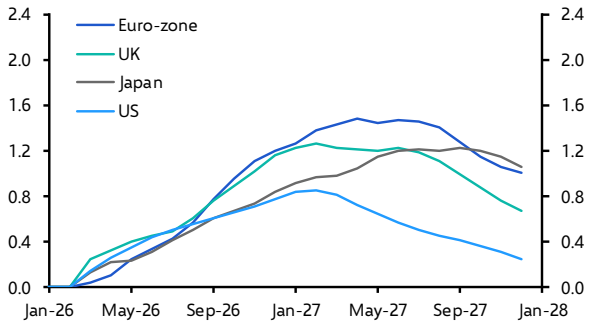
Chart 9: CPI Weights of Categories with Biggest Indirect Exposure to Higher Oil & Gas Prices (%)



Sources: LSEG, Capital Economics

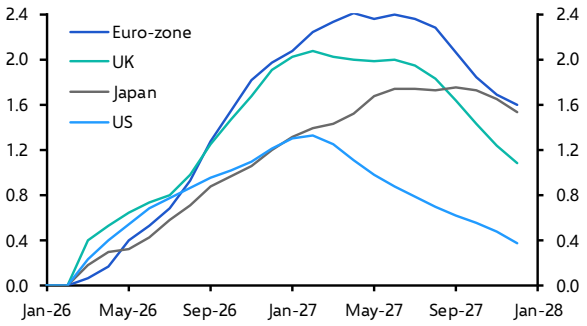
- In our baseline scenario – with Brent ending the year at $80pb and TTF natural gas at €70/MWh – **our model suggests the US will see the smallest indirect boost to inflation among major DMs**, peaking at about 0.75%-pt in early 2027. (See Chart 10.) The peak effect in the euro-zone is twice the size and doesn't unwind until H2 2027. After all, the US is mainly facing an oil price shock which fades more quickly, while in the euro-zone the far bigger rise in natural gas prices takes time to boost food and hospitality inflation. The indirect effects in the UK and Japan are a bit smaller than in the euro-zone. But because our model doesn't take policy intervention into account, in practice, fuel price caps will limit the passthrough in Japan. Price caps aside, **half or more of the indirect effects come through food and electricity, rather than core, components**.
- In our adverse scenario, with Brent reaching $130pb and TTF at €120/MWh, the indirect boosts to inflation are roughly two-thirds larger. (See Chart 11.) All else equal, **the bigger the indirect effects on inflation, the greater the risk of second-round effects taking hold and central banks acting forcefully in response**.

Chart 10: Modelled %-Point Indirect Boost to Headline Inflation in CE *Baseline* Scenario for Oil & Gas Prices



Sources: LSEG, Capital Economics

Chart 11: Modelled %-Point Indirect Boost to Headline Inflation in CE *Adverse* Scenario for Oil & Gas Prices



Sources: LSEG, Capital Economics

[1] We use a VAR framework to model the response of CPI components to shocks in oil and gas prices, controlling for shocks in other commodity prices (based on the IMF's primary commodity price index) , exchange rates, domestic economic conditions (proxied by OECD CLIs), and global supply chain conditions (proxied by S&P Global PMI indicators). Estimations based on monthly data from 1998 to 2025.

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